

PROYECTO FINAL(RENDIMIENTO ESCOLAR)

INTRODUCCIÓN A LA CIENCIA DE DATOS Y A LA COMUNIDAD DE
TECNOLOCHICAS PRO

NOMBRE DE LA ALUMNA: DENISSE BENAVIDEZ DOMÍNGUEZ



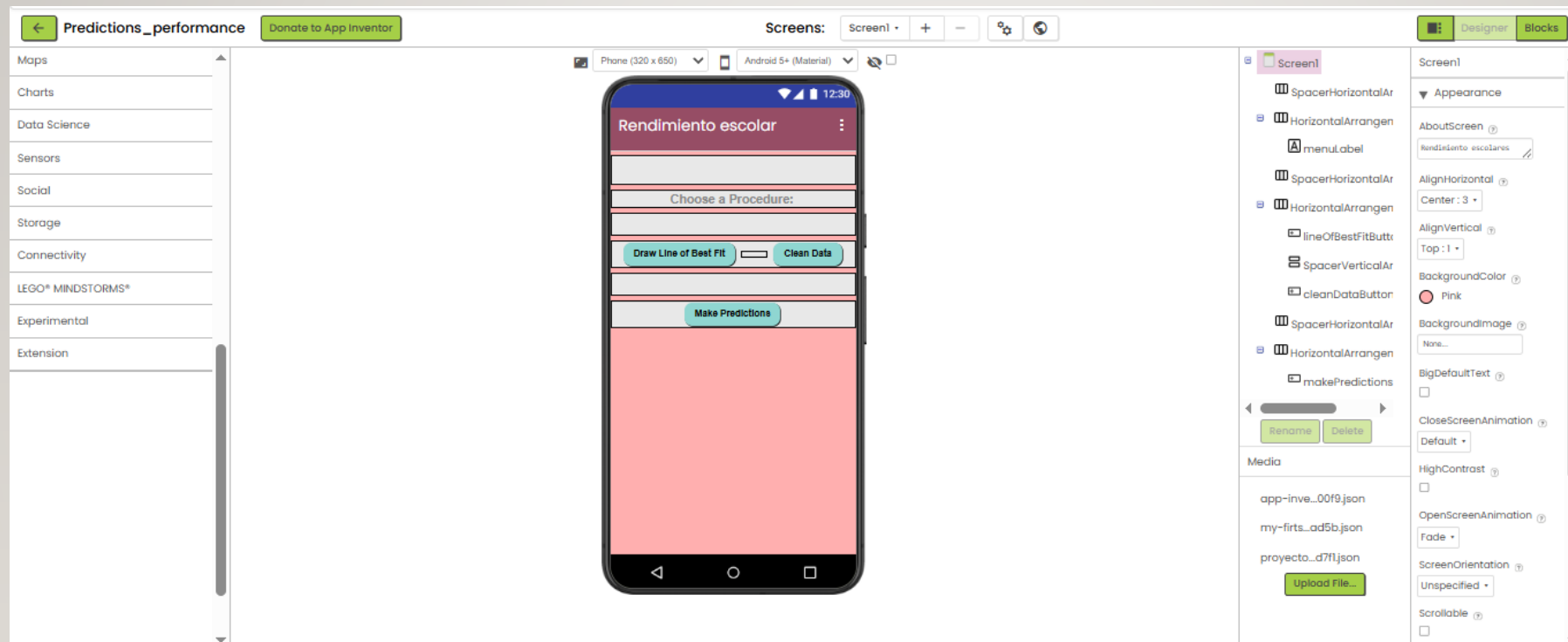
DESCRIPCIÓN

- Es una aplicación de ciencia de datos desarrollada en MIT App Inventor, permite analizar la relación en el rendimiento académico de los estudiantes, específicamente su calificación en el examen final. La aplicación sigue un flujo de trabajo de cuatro etapas y está diseñada para ser utilizada por docentes, investigadores o estudiantes interesados en comprender cómo factores como las horas de estudio influyen en el desempeño académico. Además de utilizar inteligencia artificial y una base de datos por medio de Excel.

LINKS DE GOOGLESHEETS Y OLLAMA

- https://docs.google.com/spreadsheets/d/Ip5d_NHJpPBpy6v2-DP89JK5NHjIMb_llourByhCxaKA/edit?gid=0#gid=0
- A0ead472c9b04857873d4277daa9334e

INTERFAZ I

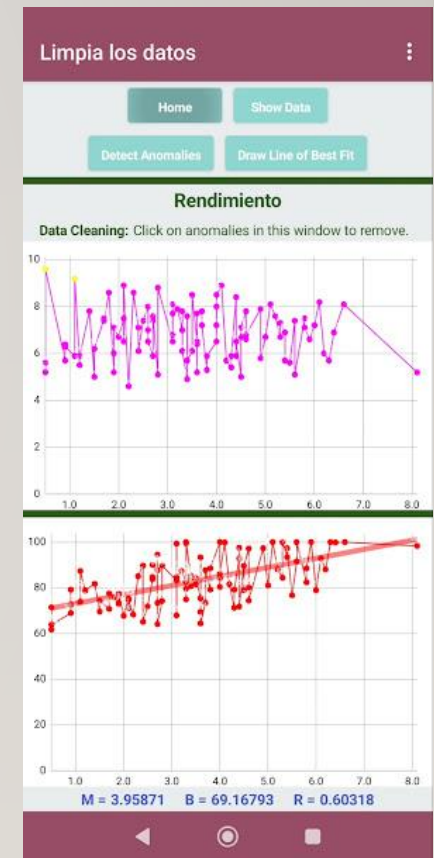
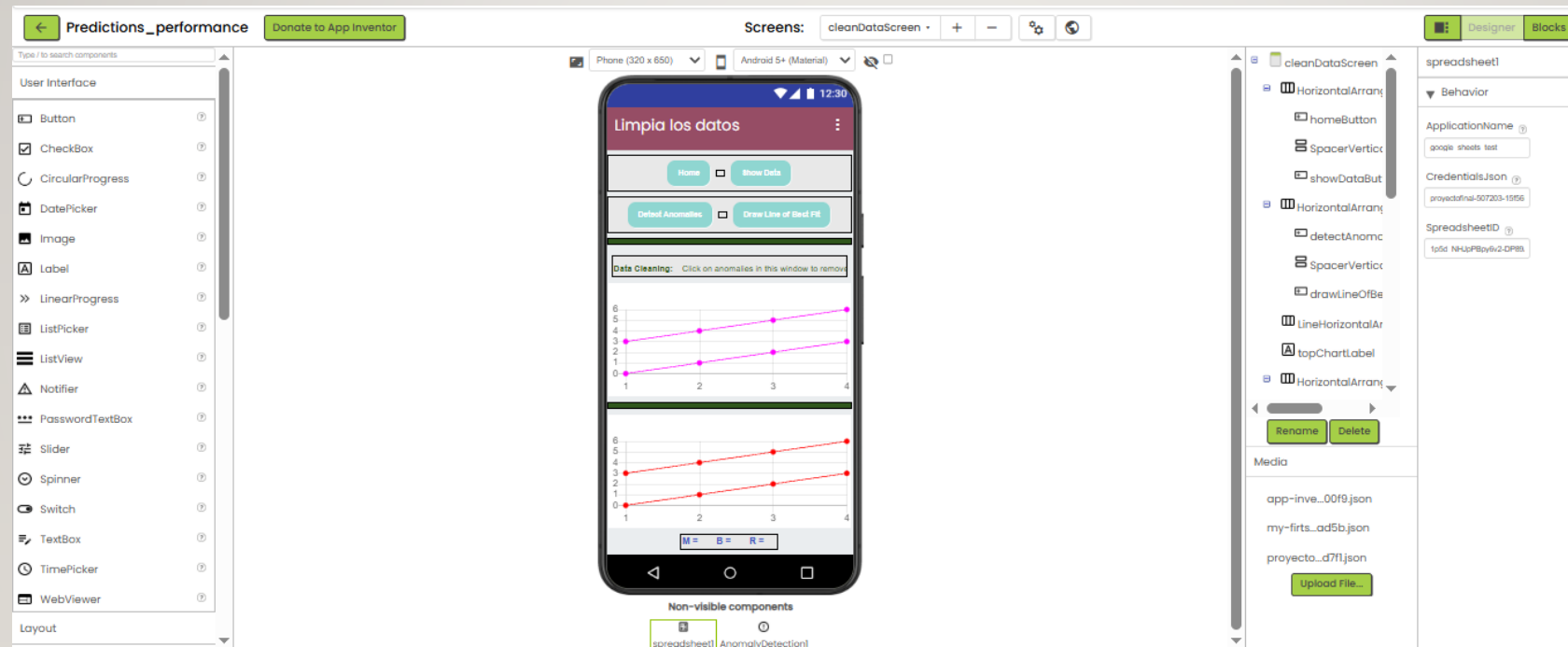


BLOQUES

The screenshot displays the App Inventor web interface for a project named "Predictions_performance". The interface is divided into several sections:

- Top Bar:** Includes a back arrow, the project name "Predictions_performance", a "Donate to App Inventor" button, and a "Screens:" section with a dropdown menu showing "Screen1" and buttons for adding (+) and removing (-) screens. On the right, there are icons for a palette, "Designer", and "Blocks" views.
- Left Panel (Built-in Palette):** A vertical list of categories for building blocks: Control, Logic, Math, Matrices, Text, Lists, Dictionaries, Colors, Variables, Procedures, and Screen1. Below this list are "Rename" and "Delete" buttons.
- Bottom Left (Media):** A section for uploading media files, showing a list of files: "app-inve...00f9.json", "my-firts...ad5b.json", and "proyecto...d7f1.json". There is an "Upload File..." button and a "Show Warnings" button with warning and error icons.
- Main Canvas (Blocks):** The central area where code blocks are assembled. It contains three event-driven code blocks:
 - Block 1:** A "when lineOfBestFitButton .Click" block followed by a "do open another screen screenName drawLOBFscreen" block.
 - Block 2:** A "when cleanDataButton .Click" block followed by a "do open another screen screenName cleanDataScreen" block.
 - Block 3:** A "when makePredictionsButton .Click" block followed by a "do open another screen screenName makePredictionsScreen" block.
- Right Panel:** Contains a vertical toolbar with icons for a backpack, a checklist, a target, zoom in (+) and zoom out (-) buttons, and a trash can.

INTERFAZ 2



BLOQUES

Predictions_performance [Donate to App Inventor](#)

Screens: cleanDataScreen + -

Designer Blocks

Built-in

- Control
- Logic
- Math
- Matrices
- Text
- Lists
- Dictionaries
- Colors
- Variables
- Procedures

cleanDataScreen

[Rename](#) [Delete](#)

Media

- app-inve...00f9.json
- my-firts...ad5b.json
- proyecto...d7f1.json
- [Upload File...](#)

when homeButton.Click

- do open another screen screenName Screen1

when showDataButton.Click

- do call cleanedChartData2D.Clear
- call dataCleaningChartData2D.Clear
- call spreadsheet1.ReadSheet
- sheetName Hoja 1

when spreadsheet1.GotSheetData

- do sheetData
- set topChartLabel.Text to Rendimiento
- call cleanedChartData2D.ImportFromSpreadsheet
- spreadsheet spreadsheet1
- xColumn Tiempo de estudio
- yColumn Ultima calificación
- useHeaders true
- call dataCleaningChartData2D.ImportFromSpreadsheet
- spreadsheet spreadsheet1
- xColumn Tiempo de estudio
- yColumn Horas de sueño
- useHeaders true

when drawLineOfBestFitButton.Click

- do set Trendline1.ChartData to cleanedChartData2D

when detectAnomaliesButton.Click

- do call dataCleaningChartData2D.HighlightDataPoints
- dataPoints call AnomalyDetection1.DetectAnomaliesInChartData
- chartData dataCleaningChartData2D
- threshold 2
- color yellow

when dataCleaningChartData2D.EntryClick

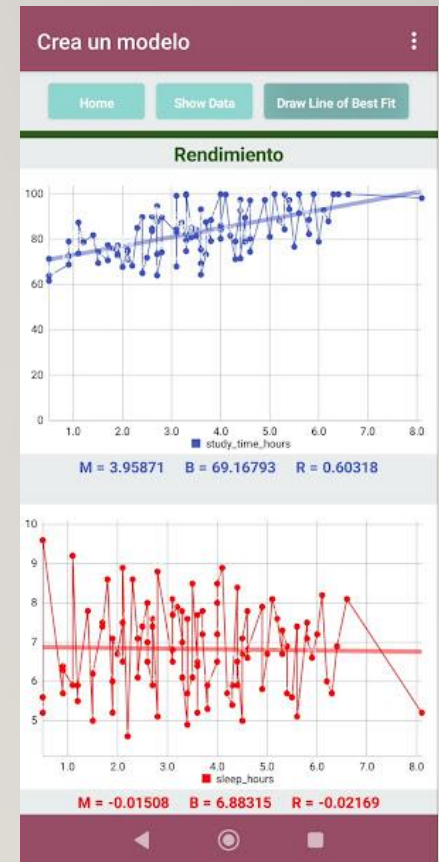
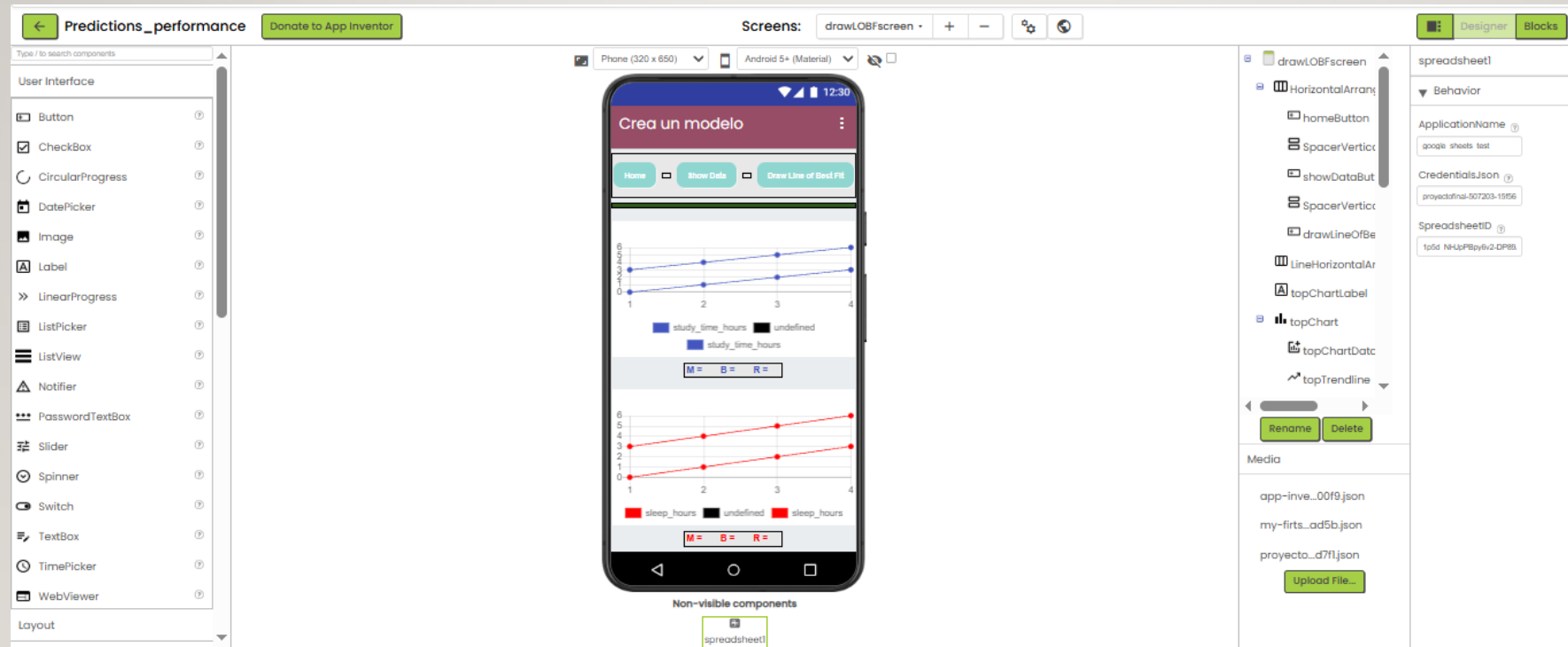
- do if is in list? thing make a list
- get x
- get y
- list call AnomalyDetection1.DetectAnomaliesInChartData
- chartData dataCleaningChartData2D
- threshold 2
- then call dataCleaningChartData2D.RemoveEntry
- x get x
- y get y
- call cleanedChartData2D.Clear
- call cleanedChartData2D.ImportFromList
- list call dataCleaningChartData2D.GetAllEntries

when Trendline1.Updated

- do results
- set topSlopeValueLabel.Text to Trendline1.LinearCoefficient
- set topY_intValueLabel.Text to Trendline1.YIntercept
- set topCor_coefValueLabel.Text to Trendline1.CorrelationCoefficient

Show Warnings

INTERFAZ 3

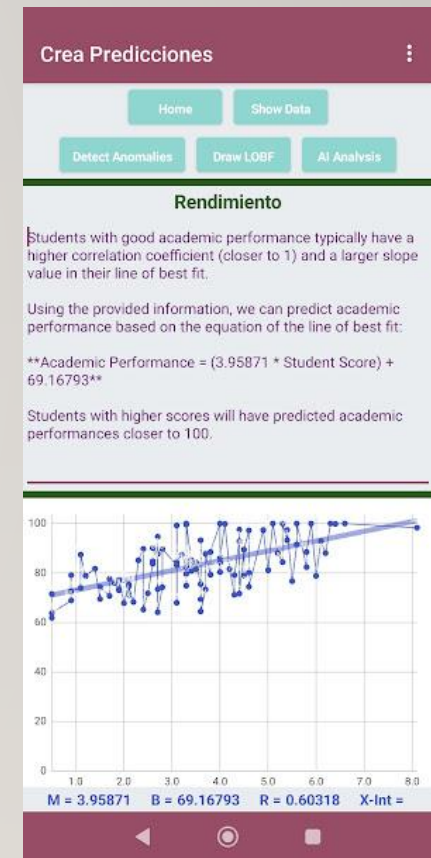
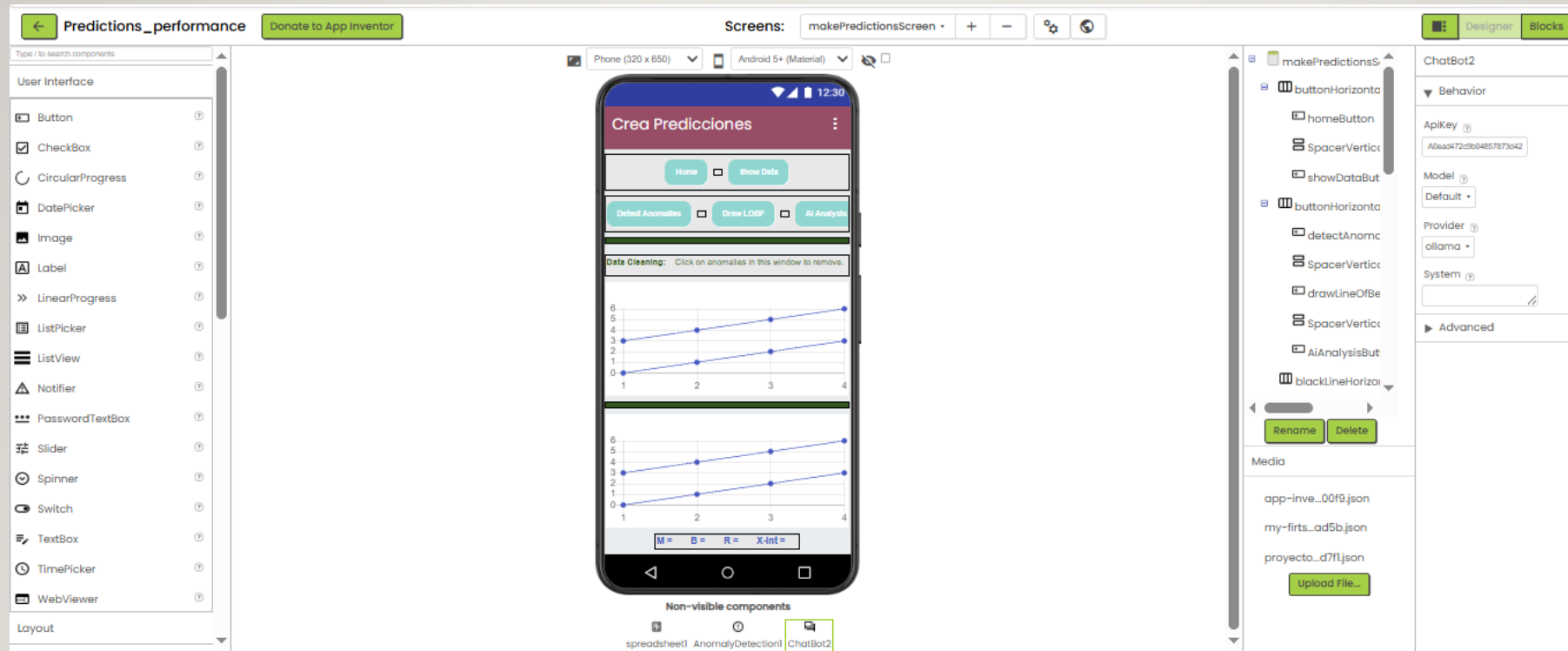


BLOQUES

The screenshot displays the App Inventor web interface for a project titled "Predictions_performance". The interface is divided into several sections:

- Top Bar:** Includes a back arrow, the project name "Predictions_performance", a "Donate to App Inventor" button, and a "Screens:" dropdown menu currently set to "drawLOBFscreen".
- Left Panel:** Contains a "Built-in" category with various block types (Control, Logic, Math, Matrices, Text, Lists, Dictionaries, Colors, Variables, Procedures) and a "Media" section with file upload options.
- Main Canvas:** Shows a visual programming block for the "drawLOBFscreen" screen. The block is structured as follows:
 - when homeButton.Click** event triggers the **do** block: **open another screen** with screenName "Screen1".
 - when showDataButton.Click** event triggers the **do** block:
 - call topChartData2D.Clear**
 - call bottomChartData2D.Clear**
 - call spreadsheet1.ReadSheet** with sheetName "Hoja 1".
 - when spreadsheet1.GotSheetData** event triggers the **do** block:
 - set topChartLabel.Text** to "Rendimiento".
 - call topChartData2D.ImportFromSpreadsheet** with parameters: spreadsheet "spreadsheet1", xColumn "Tiempo de estudio", yColumn "Ultima calificación", and useHeaders "true".
 - call bottomChartData2D.ImportFromSpreadsheet** with parameters: spreadsheet "spreadsheet1", xColumn "Tiempo de estudio", yColumn "Horas de sueño", and useHeaders "true".
 - when drawLineOfBestFitButton.Click** event triggers the **do** block:
 - set topTrendline.ChartData** to **topChartData2D**.
 - set bottomTrendline.ChartData** to **bottomChartData2D**.
 - set topSlopeValueLabel.Text** to **topTrendline.LinearCoefficient**.
 - set topY_intValueLabel.Text** to **topTrendline.YIntercept**.
 - set topCor_coefValueLabel.Text** to **topTrendline.CorrelationCoefficient**.
 - set bottomSlopeValueLabel.Text** to **bottomTrendline.LinearCoefficient**.
 - set bottomY_intValueLabel.Text** to **bottomTrendline.YIntercept**.
 - set bottomCor_coefValueLabel.Text** to **bottomTrendline.CorrelationCoefficient**.
- Right Panel:** Includes a "Designer" tab, a "Blocks" tab, and a "Show Warnings" button at the bottom.

INTERFAZ 4



BLOQUES

The screenshot displays the App Inventor web interface for a project named "Predictions_performance". The interface is divided into several sections:

- Top Bar:** Includes a back arrow, the project name "Predictions_performance", a "Donate to App Inventor" button, and a "Screens:" section showing "makePredictionsScreen" with plus and minus icons. On the right, there are icons for "Designer" and "Blocks", and a backpack icon.
- Left Panel (Built-in):** A vertical menu with categories: Control, Logic, Math, Matrices, Text, Lists, Dictionaries, Colors, Variables, and Procedures. Below these are "makePredictionsScreen" and "Media" sections. The "Media" section lists files: "app-inve...00f9.json", "my-firts...ad5b.json", and "proyecto...d7f1.json", with an "Upload File..." button.
- Main Canvas (Code Blocks):**
 - when homeButton.Click**: do open another screen screenName Screen1
 - when showDataButton.Click**: do call cleanedChartData2D.Clear, call dataCleaningChartData2D.Clear, call spreadsheet1.ReadSheet sheetName Hoja 1
 - when spreadsheet1.GotSheetData**: sheetData do set topChartLabel.Text to Rendimiento, call cleanedChartData2D.ImportFromSpreadsheet spreadsheet spreadsheet1, xColumn Tiempo de estudio, yColumn Ultima calificacion, useHeaders true, call dataCleaningChartData2D.ImportFromSpreadsheet spreadsheet spreadsheet1, xColumn Tiempo de estudio, yColumn Horas de sueño, useHeaders true
 - when drawLineOfBestFitButton.Click**: do set Trendline1.ChartData to cleanedChartData2D
 - when Trendline1.Updated**: results
 - when ChatBot2.GotResponse**: responseText do set AiResponseTextBox.Text to get responseText
 - when AiAnalysisButton.Click**: do set dataCleaningChart.Visible to false, set dataCleaningHorizontalArrangement.Visible to false, set AiResponseHorizontalArrangement.Visible to true, call ChatBot2.Converse question join
 - Given the following data
 - How many students have good academic performance?:
 - call cleanedChartData2D.GetAllEntries
 - The correlation coefficient for the line of best...
 - Trendline1.CorrelationCoefficient
 - The slope of the line of best fit is
 - Trendline1.LinearCoefficient
 - The y-intercept for the line of best fit is
 - Trendline1.YIntercept
 - Predict the academic performance of these studen...
 - Limit your response to 120 words.

- Right Panel:** Contains a vertical scrollbar and a trash can icon.

BLOQUES

Predictions_performance [Donate to App Inventor](#)

Screens: **makePredictionsScreen** + -

Designer **Blocks**

Built-in

- Control
- Logic
- Math
- Matrices
- Text
- Lists
- Dictionaries
- Colors
- Variables
- Procedures

makePredictionsScreen

[Rename](#) [Delete](#)

Media

- app-inve...00f9.json
- my-firts...ad5b.json
- proyecto...d7f1.json

[Upload File...](#)

Code:

```
when drawLineOfBestFitButton.Click
do
  set Trendline1.ChartData to cleanedChartData2D

when Trendline1.Updated
do
  set SlopeValueLabel.Text to Trendline1.LinearCoefficient
  set Y_intValueLabel.Text to Trendline1.YIntercept
  set Cor_coefValueLabel.Text to Trendline1.CorrelationCoefficient

when detectAnomaliesButton.Click
do
  call dataCleaningChartData2D.HighlightDataPoints
  dataPoints call AnomalyDetection1.DetectAnomaliesInChartData
  chartData dataCleaningChartData2D
  threshold 2
  color yellow

when dataCleaningChartData2D.EntryClick
do
  call dataCleaningChartData2D.RemoveEntry
  x get x
  y get y
  call cleanedChartData2D.Clear
  call cleanedChartData2D.ImportFromList
  list call dataCleaningChartData2D.GetAllEntries
```

Show Warnings